

MH1101 Tutorial 7 (Week 8) Solution

Reference: Sections 4.1, 4.2 (Lecture Notes)

1. Using the formal definition of limit, prove that the following sequence converges.

(i) $\left\{ \frac{1}{3^n} \right\}_{n=1}^{\infty}$

(ii) $\left\{ \frac{2n^2 + 3}{n^2 + 1} \right\}_{n=1}^{\infty}$

(iii) $\lim_{n \rightarrow \infty} \frac{2n^2 - 3n}{3n^2 + 1} = \frac{2}{3}$. [AY2021 Sem 2 EXAM]

Solution

(i) We will prove that the sequence converges to 0.

Let $\varepsilon > 0$. We want to find an integer N such that

$$n > N \implies \left| \frac{1}{3^n} - 0 \right| < \varepsilon.$$

Working backwards to find the N :

$$\begin{aligned} \left| \frac{1}{3^n} - 0 \right| < \varepsilon &\iff \frac{1}{3^n} < \varepsilon \\ &\iff 3^n > \varepsilon^{-1} \\ &\iff n \ln 3 > \ln \varepsilon^{-1} \\ &\iff n > \frac{\ln \varepsilon^{-1}}{\ln 3}. \end{aligned} \tag{1}$$

We choose an integer $N = \left\lceil \frac{\ln \varepsilon^{-1}}{\ln 3} \right\rceil$. Then

$$n > N \implies n > \left\lceil \frac{\ln \varepsilon^{-1}}{\ln 3} \right\rceil \geq \frac{\ln \varepsilon^{-1}}{\ln 3}$$

and so

$$n > N \implies \left| \frac{1}{3^n} - 0 \right| < \varepsilon,$$

as desired.

(ii) We will prove that the sequence converges to 2.

Let $\varepsilon > 0$. We want to find an integer N such that

$$n > N \implies \left| \frac{2n^2 + 3}{n^2 + 1} - 2 \right| < \varepsilon.$$

We first make a simplification:

$$\frac{2n^2 + 3}{n^2 + 1} - 2 = \frac{2n^2 + 3 - 2n^2 - 2}{n^2 + 1} = \frac{1}{n^2 + 1}$$

Working backwards to find the N :

$$\begin{aligned} \left| \frac{2n^2 + 3}{n^2 + 1} - 2 \right| < \varepsilon &\iff \frac{1}{n^2 + 1} < \varepsilon \\ &\iff n^2 + 1 > \varepsilon^{-1} \\ &\iff n^2 > \varepsilon^{-1} - 1. \end{aligned} \tag{2}$$

We choose an integer $N = \left\lceil \sqrt{|\varepsilon^{-1} - 1|} \right\rceil$. Then

$$\begin{aligned} n > N &\implies n > \left\lceil \sqrt{|\varepsilon^{-1} - 1|} \right\rceil \\ &\implies n > \sqrt{|\varepsilon^{-1} - 1|} \implies n^2 > |\varepsilon^{-1} - 1| \\ &\implies n^2 > \varepsilon^{-1} - 1. \end{aligned}$$

Thus, for $n > N$, the relation (4) holds, and so

$$n > N \implies \left| \frac{2n^2 + 3}{n^2 + 1} - 2 \right| < \varepsilon,$$

as desired.

(iii) Let $\epsilon > 0$.

Note that

$$\begin{aligned} \left| \frac{2n^2 - 3n}{3n^2 + 1} - \frac{2}{3} \right| &= \left| \frac{3(2n^2 - 3n) - 2(3n^2 + 1)}{3(3n^2 + 1)} \right| = \left| \frac{-9n - 2}{3(3n^2 + 1)} \right| \\ &= \left| \frac{9n + 2}{3(3n^2 + 1)} \right| \\ &\leq \frac{9n + 2}{3(3n^2)} \\ &\leq \frac{9n + n}{3(3n^2)} \quad \text{if } n \geq 2 \quad \text{---(1)} \\ &\leq \frac{10}{9n} < \epsilon \quad \text{if } n > \frac{10}{9\epsilon} \quad \text{---(2)} \end{aligned}$$

Let $N = \max(2, \lceil \frac{10}{9\epsilon} \rceil)$. If $n > N$, then $n > 2$ and $n > \frac{10}{9\epsilon}$. It follows from (1) and (2) that

$$\left| \frac{2n^2 - 3n}{3n^2 + 1} - \frac{2}{3} \right| < \epsilon.$$

This proves

$$\lim_{n \rightarrow \infty} \frac{2n^2 - 3n}{3n^2 + 1} = \frac{2}{3}.$$

□

2. Determine whether the sequence converges or diverges. If it converges, find the limit. Justify your answer.

$$(i) \ a_n = \frac{3 + 5n^2}{n + n^2}$$

$$(vi) \ a_n = \frac{4^n}{1 + 9^n}$$

$$(ii) \ a_n = \frac{n^4}{n^3 - 2n}$$

$$(vii) \ a_n = \frac{(-1)^n n^3}{n^3 + 2n^2 + 1}$$

$$(iii) \ a_n = \sqrt{\frac{n+1}{9n+1}}$$

$$(viii) \ a_n = \tan\left(\frac{2n\pi}{1+8n}\right)$$

$$(iv) \ a_n = \frac{\cos^2 n}{2^n}$$

$$(ix) \ a_n = \frac{\tan^{-1} n}{n}$$

$$(v) \ a_n = \sqrt[n]{2^{1+3n}}$$

$$(x) \ a_n = \sqrt[n]{n}$$

Solution

(i) Divide every term by n^2 :

$$a_n = \frac{3 + 5n^2}{n + n^2} = \frac{\frac{3}{n^2} + 5}{\frac{1}{n} + 1}$$

Since $\frac{3}{n^2} \rightarrow 0$ and $\frac{1}{n} \rightarrow 0$, using Limit Laws, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \frac{\lim_{n \rightarrow \infty} \frac{3}{n^2} + 5}{\lim_{n \rightarrow \infty} \frac{1}{n} + 1} \\ &= \frac{0 + 5}{0 + 1} = 5. \end{aligned}$$

The sequences converges to 5.

(ii) Dividing each term by the dominant term n^4 :

$$\lim_{n \rightarrow \infty} \frac{n^4}{n^3 - 2n} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n} - \frac{2}{n^3}}$$

Since $\frac{1}{n} - \frac{2}{n^3}$ for $n \geq 2$, we have $\lim_{n \rightarrow \infty} \frac{1}{n} - \frac{2}{n^3} = 0^+$ ($\frac{1}{n} - \frac{2}{n^3}$ approaching 0 from the right). Thus

$$\lim_{n \rightarrow \infty} \frac{n^4}{n^3 - 2n} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n} - \frac{2}{n^3}} = \infty.$$

(iii) Using Limit Laws and the continuity of the square root function,

$$\begin{aligned}
 \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \sqrt{\frac{n+1}{9n+1}} \\
 &= \sqrt{\lim_{n \rightarrow \infty} \frac{n+1}{9n+1}} \\
 &= \sqrt{\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{9 + \frac{1}{n}}} \\
 &= \sqrt{\frac{1 + \lim_{n \rightarrow \infty} \frac{1}{n}}{9 + \lim_{n \rightarrow \infty} \frac{1}{n}}} \\
 &= \sqrt{\frac{1+0}{9+0}} \\
 &= \frac{1}{3}.
 \end{aligned}$$

The sequence converges to $\frac{1}{3}$.

(iv) Since $0 \leq \cos^2 n \leq 1$, we have

$$0 \leq \frac{\cos^2 n}{2^n} \leq \frac{1}{2^n}.$$

Note that

$$\lim_{n \rightarrow \infty} \frac{1}{2^n} = \lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0.$$

So, by the Squeeze Theorem, the sequence $\{a_n\} = \left\{\frac{\cos^2 n}{2^n}\right\}$ converges to 0.

(v) Write

$$a_n = \sqrt[n]{2^{1+3n}} = 2^{\frac{1+3n}{n}} = (2 \cdot 2^{3n})^{1/n} = 2^{1/n} 2^3 = 8 \cdot 2^{1/n}.$$

The function $f(x) = 2^x$ is continuous at 0.

Since $\frac{1}{n} \rightarrow 0$, by the continuity of f , we have $\lim_{n \rightarrow \infty} 2^{1/n} = 2^{\lim_{n \rightarrow \infty} 1/n} = 2^0 = 1$.

Hence,

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} 8 \cdot 2^{1/n} \\ &= 8 \cdot \lim_{n \rightarrow \infty} 2^{1/n} \\ &= 8 \cdot 1 = 8.\end{aligned}$$

The sequence converges to 8.

(vi) Divide every term by 9^n :

$$a_n = \frac{4^n}{1 + 9^n} = \frac{\left(\frac{4}{9}\right)^n}{\left(\frac{1}{9}\right)^n + 1}$$

Recall that $\lim_{n \rightarrow \infty} r^n = 0$ for $-1 < r < 1$. So, by Limit Laws,

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \frac{\lim_{n \rightarrow \infty} \left(\frac{4}{9}\right)^n}{\lim_{n \rightarrow \infty} \left(\frac{1}{9}\right)^n + 1} \\ &= \frac{0}{0 + 1} = 0.\end{aligned}$$

The sequence converges to 0.

Alternatively, we can apply Squeeze Theorem:

$$0 \leq a_n = \frac{4^n}{1 + 9^n} \leq \frac{4^n}{9^n} = \left(\frac{4}{9}\right)^n.$$

Since $\lim_{n \rightarrow \infty} \left(\frac{4}{9}\right)^n = 0$, we deduce from Squeeze Theorem that $\lim_{n \rightarrow \infty} a_n = 0$.

(vii) Consider the even subsequences $\{a_2, a_4, a_6, \dots\}$ and the odd subsequences $\{a_1, a_3, a_5, \dots\}$.

Let $b_n = \frac{n^3}{n^3 + 2n^2 + 1}$, and so we can write $a_n = (-1)^n b_n$.

Dividing every term by n^3 :

$$b_n = \frac{n^3}{n^3 + 2n^2 + 1} = \frac{1}{1 + \frac{2}{n} + \frac{1}{n^3}}.$$

By Limit Laws, we have

$$\lim_{n \rightarrow \infty} b_n = \frac{1}{1 + \lim_{n \rightarrow \infty} \frac{2}{n} + \lim_{n \rightarrow \infty} \frac{1}{n^3}} = \frac{1}{1 + 0 + 0} = 1.$$

Therefore,

- If n is even, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^n b_n = \lim_{n \rightarrow \infty} b_n = 1$.
- If n is odd, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^n b_n = \lim_{n \rightarrow \infty} -b_n = -\lim_{n \rightarrow \infty} b_n = -1$.

The even and odd subsequences converge to different limits, so by the Subsequence Test, the given sequence $\{a_n\}$ is divergent.

(viii) Let $b_n = \frac{2n\pi}{1+8n}$. Then

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{2\pi}{\frac{1}{n} + 8} = \frac{2\pi}{\lim_{n \rightarrow \infty} \frac{1}{n} + 8} = \frac{2\pi}{0 + 8} = \frac{\pi}{4}.$$

Since $\tan$ function is continuous at $\frac{\pi}{4}$, by the continuity of f ,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \tan\left(\frac{2n\pi}{1+8n}\right) = \tan\left(\lim_{n \rightarrow \infty} \frac{2n\pi}{1+8n}\right) = \tan\left(\frac{\pi}{4}\right) = 1.$$

The sequence $\{a_n\}$ converges to 1.

(ix) Let $b_n = \tan^{-1} n$. The sequence $\{b_n\}$ is defined by the function $f(x) = \tan^{-1}(x)$. Also,

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}.$$

So by Theorem 2 (Sequence defined by a function),

$$\lim_{n \rightarrow \infty} b_n = \lim_{x \rightarrow \infty} f(x) = \frac{\pi}{2}.$$

On the other hand, let $c_n = \frac{1}{n}$. Clearly,

$$\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Hence,

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} b_n c_n \\ &= \left(\lim_{n \rightarrow \infty} b_n\right) \cdot \left(\lim_{n \rightarrow \infty} c_n\right) \\ &= \frac{\pi}{2} \cdot 0 \\ &= 0. \end{aligned}$$

Alternatively, we can apply Squeeze Theorem:

$$-\frac{\pi/2}{n} \leq a_n = \frac{\tan^{-1} n}{n} \leq \frac{\pi/2}{n}.$$

Since $\lim_{n \rightarrow \infty} \frac{\pm\pi/2}{n} = 0$, we deduce from Squeeze Theorem that $\lim_{n \rightarrow \infty} a_n = 0$.

(x) Let $b_n = \ln a_n = \ln(n^{1/n}) = \frac{\ln n}{n}$. We can write

$$a_n = e^{\ln a_n} = e^{b_n}.$$

Consider the function $f(x) = \frac{\ln x}{x}$. By L'Hopital's rule,

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

By Theorem 2 (Sequence defined by a function),

$$\lim_{n \rightarrow \infty} b_n = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0.$$

Consider the function $g(f(x)) = e^{f(x)} = e^{\frac{\ln x}{x}}$, where $g(x) = e^x$. By the continuity of $g(x)$ at 0, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} e^{b_n} \\ &= e^{\lim_{n \rightarrow \infty} b_n} \\ &= e^0 = 1. \end{aligned}$$

The sequence $\{a_n\}$ converges to 1.

3. Given a sequence $\{a_n\}_{n=1}^{\infty}$, prove that (using the formal definition of limit) if the odd subsequence $a_1, a_3, \dots$ and the even subsequence $a_2, a_4, \dots$ both converge to the same limit L , then $\{a_n\}$ converges to the limit L .

Solution Let $\varepsilon > 0$. We want to find a positive integer N such that

$$n > N \implies |a_n - L| < \varepsilon.$$

Since the odd subsequence $\{a_{2m-1}\}_{m=1}^{\infty}$ converges to L , there exists positive integer N_1 such that

$$m > N_1 \implies |a_{2m-1} - L| < \varepsilon. \quad (3)$$

Likewise, since the even subsequence $\{a_{2m}\}_{m=1}^{\infty}$ converges to L , there exists positive integer N_2 such that

$$m > N_2 \implies |a_{2m} - L| < \varepsilon. \quad (4)$$

Set $N = \max\{2N_1 - 1, 2N_2\}$. Suppose $n > N$.

- If n is odd, say $n = 2m - 1$ for some m . Then

$$n > N \implies 2m - 1 > \max\{2N_1 - 1, 2N_2\} \geq 2N_1 - 1 \implies m > N_1.$$

So by (3), we have $|a_n - L| < \varepsilon$.

- If n is even, say $n = 2m$ for some m . Then

$$n > N \implies 2m > \max\{2N_1 - 1, 2N_2\} \geq 2N_2 \implies m > N_2.$$

So by (4), we have $|a_n - L| < \varepsilon$.

Combining, we have proved that

$$n > N \implies |a_n - L| < \varepsilon.$$

That is, the sequence $\{a_n\}$ converges to L .

4. Determine whether the statement is true or false. If it is true, explain why it is true. If it is false, explain why it is false, or give an example to show that it is false.

- (a) If $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are divergent, then $\{a_n + b_n\}_{n=1}^{\infty}$ is divergent.
- (b) If $\{a_n\}$ is divergent, then $\{|a_n|\}$ is divergent.
- (c) If $\{a_n\}$ converges to L (real number) and $\{b_n\}$ converges to 0, the $\{a_n b_n\}$ converges to 0.

Solution

- (a) False. Take $a_n = (-1)^{n+1}$ and $b_n = (-1)^n$. Both $\{a_n\}$ and $\{b_n\}$ diverge. However, the sequence $\{a_n + b_n\}$ converges since $a_n + b_n = (-1)^{n+1} + (-1)^n = 0$ for all $n \geq 1$.
- (b) False. Let $n \geq 1$. Take $a_n = (-1)^n$. The sequence $\{a_n\}$ diverges, but the sequence $\{|a_n|\}$ converges since $|a_n| = 1$ for all $n \geq 1$.
- (c) True. This follows from the limit law that the limit of a sum is the sum of limits: $\lim_{n \rightarrow \infty} a_n b_n = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n = L \cdot 0 = 0$.

To see this from the first principle, let $\epsilon > 0$.

Since $a_n \rightarrow L$, there exists N such that $|a_n - L| < 1$ for all $n > N$ (applying the definition with 'epsilon' chosen to be 1), and so $|a_n| < |L| + 1$ for all $n > N$.

Also, since $b_n \rightarrow 0$, there exists M such that $|b_n - 0| < \frac{\epsilon}{|L|+1}$ for all $n > M$. (Here we apply the definition with 'epsilon' chosen to be $\frac{\epsilon}{|L|+1}$.)

Let $K = \max\{N, M\}$. Then for all $n > K$, we have

$$|a_n b_n - 0| = |a_n| |b_n| < (|L| + 1) \cdot \frac{\epsilon}{|L| + 1} = \epsilon.$$

That is, the sequence $\{a_n b_n\}$ converges to 0.

5. Use Squeeze Theorem to prove that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1, \quad a > 0.$$

Solution Since $a > 0$, we can always find positive integer N_1 and N_2 large enough so that $\frac{1}{N_2} < a < N_1$. Then for $n > \max\{N_1, N_2\}$,

$$\frac{1}{n} < \frac{1}{N_2} < a < N_1 < n.$$

Taking n -th root on both sides:

$$\frac{1}{n^{1/n}} < a^{1/n} < n^{1/n}.$$

Since $\lim_{n \rightarrow \infty} n^{1/n} = 1$ (see Question 2(x)), we deduce by Squeeze Theorem that $\lim_{n \rightarrow \infty} a^{1/n} = 1$. $\square$

Answer Keys. **2.** (i) 5 (ii) ∞ (iii) $\frac{1}{3}$ (iv) 0 (v) 8 (vi) 0 (vii) diverges (viii) 1 (ix) 0 (x) 1